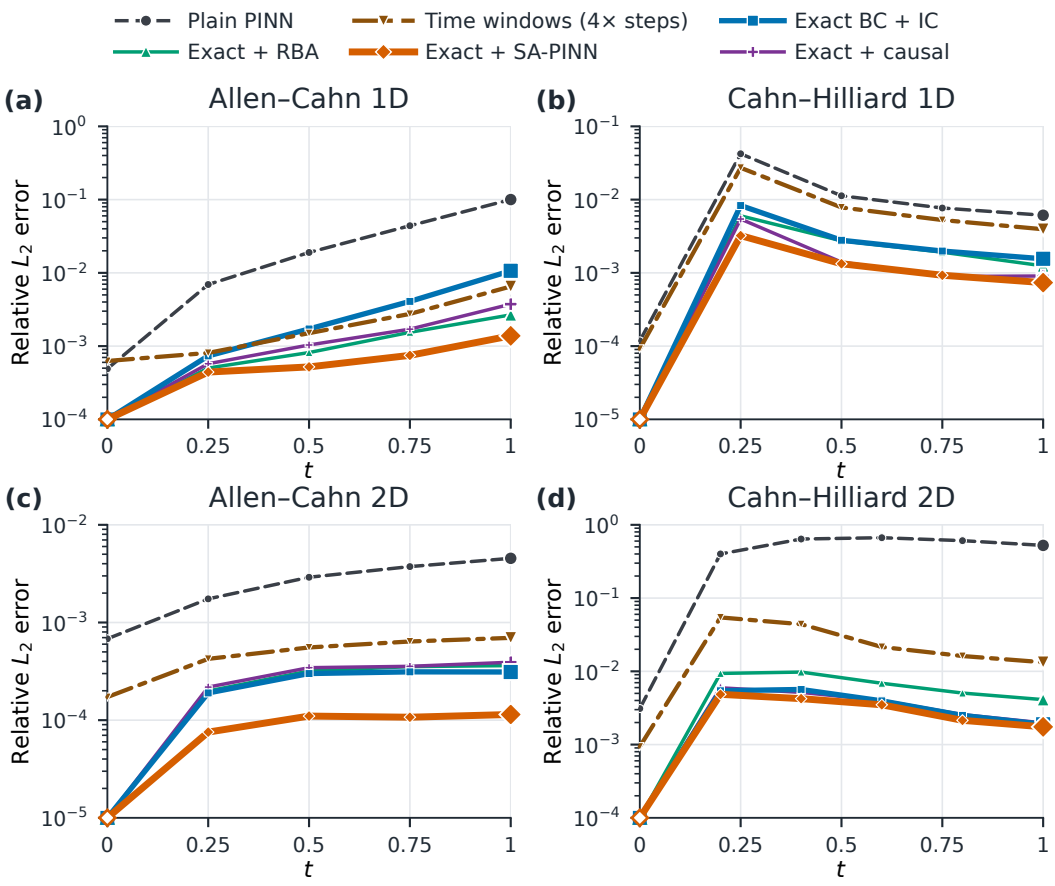
Plain PINN
Time windows (4× steps)
Exact BC + IC
Exact + RBA
Exact + SA-PINN
Exact + causal
(a)
Allen–Cahn 1D
(b)
Cahn–Hilliard 1D
(c)
Allen–Cahn 2D
(d)
Cahn–Hilliard 2D
Relative L_2 error
t
0
0.25
0.5
0.75
1
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}